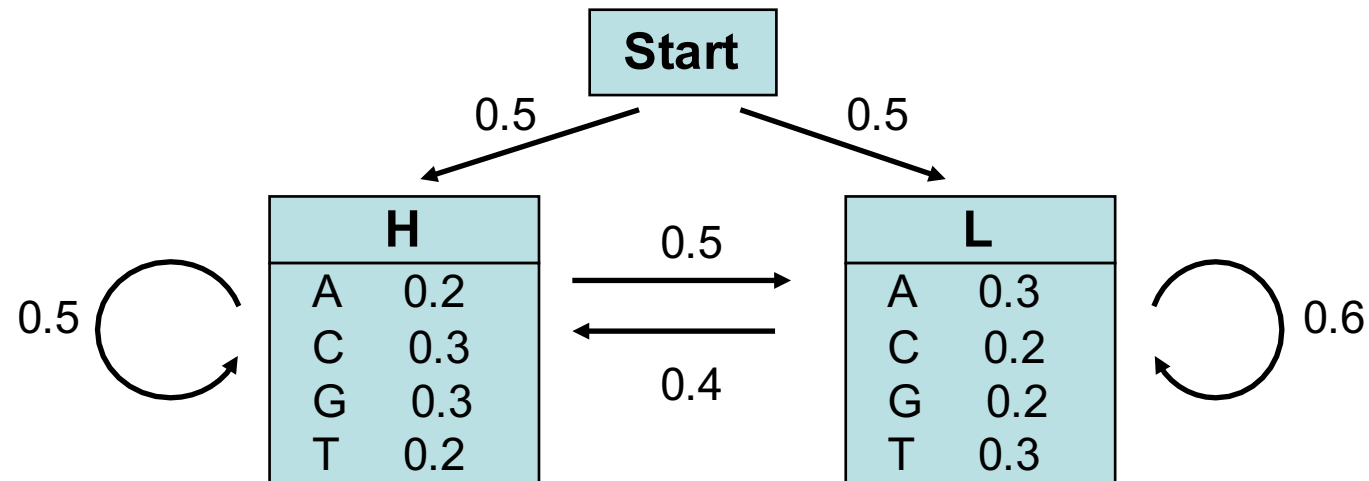


HMM : Viterbi algorithm - a toy example

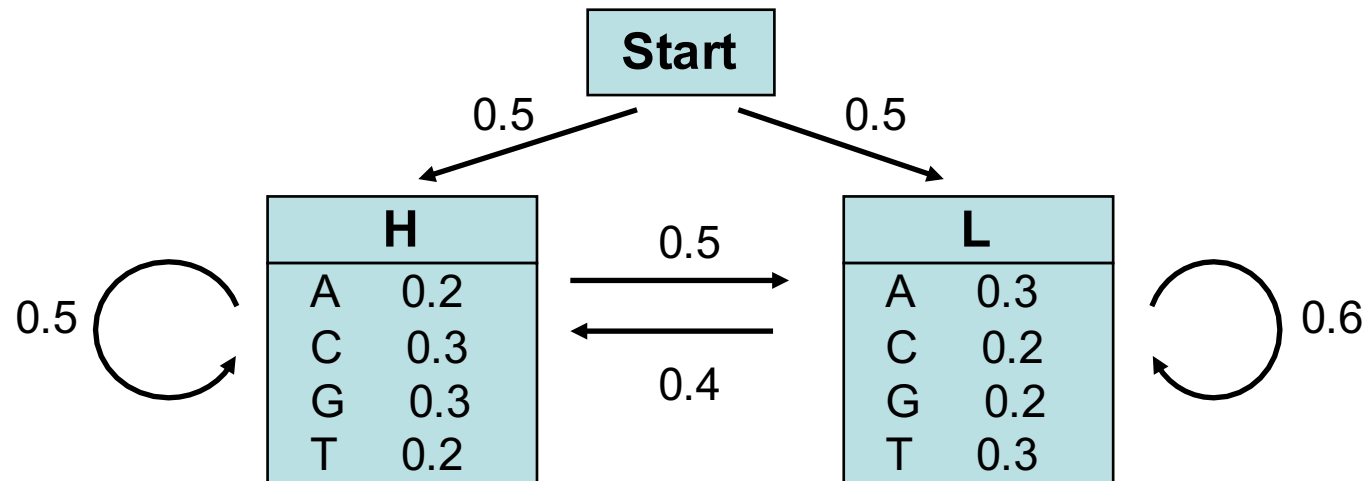


Let's consider the following simple HMM. This model is composed of 2 states, **H** (high GC content) and **L** (low GC content). We can for example consider that state H characterizes coding DNA while L characterizes non-coding DNA.

The model can then be used to predict the region of coding DNA from a given sequence.

Sources: For the theory, see Durbin *et al* (1998);
For the example, see Borodovsky & Ekisheva (2006), pp 80-81

HMM : Viterbi algorithm - a toy example



Consider the sequence $S = \text{GGCACTGAA}$

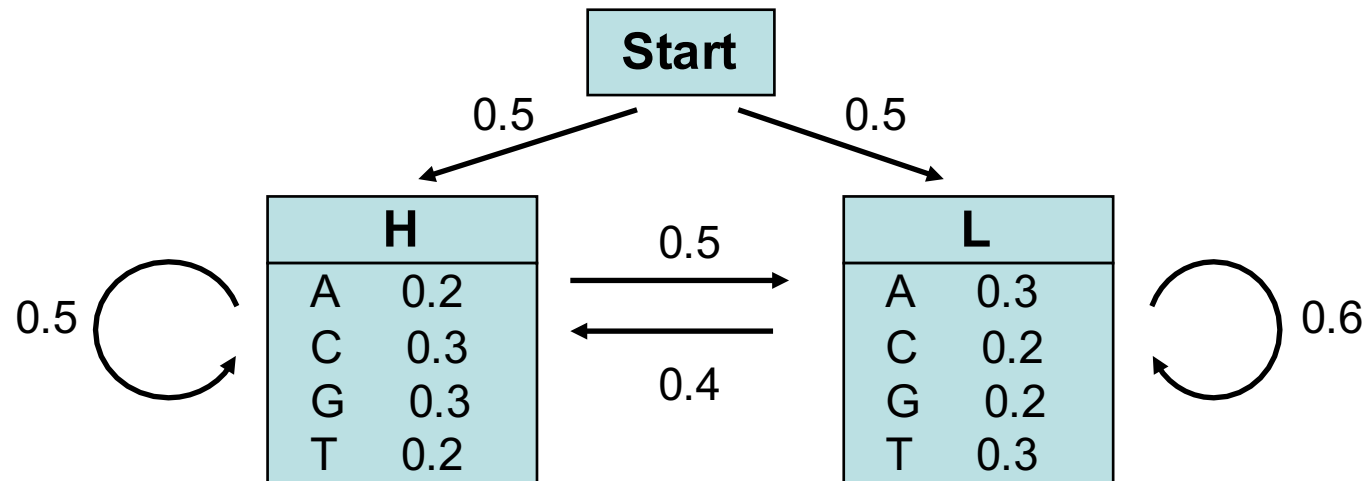
There are several paths through the hidden states (H and L) that lead to the given sequence S.

Example: $P = \text{LLHHHLLL}$

The probability of the HMM to produce sequence S through the path P is:

$$\begin{aligned} p &= p_L(0) * p_L(G) * p_{LL} * p_L(G) * p_{LH} * p_H(C) * \dots \\ &= 0.5 * 0.2 * 0.6 * 0.2 * 0.4 * 0.3 * \dots \\ &= \dots \end{aligned}$$

HMM : Viterbi algorithm - a toy example

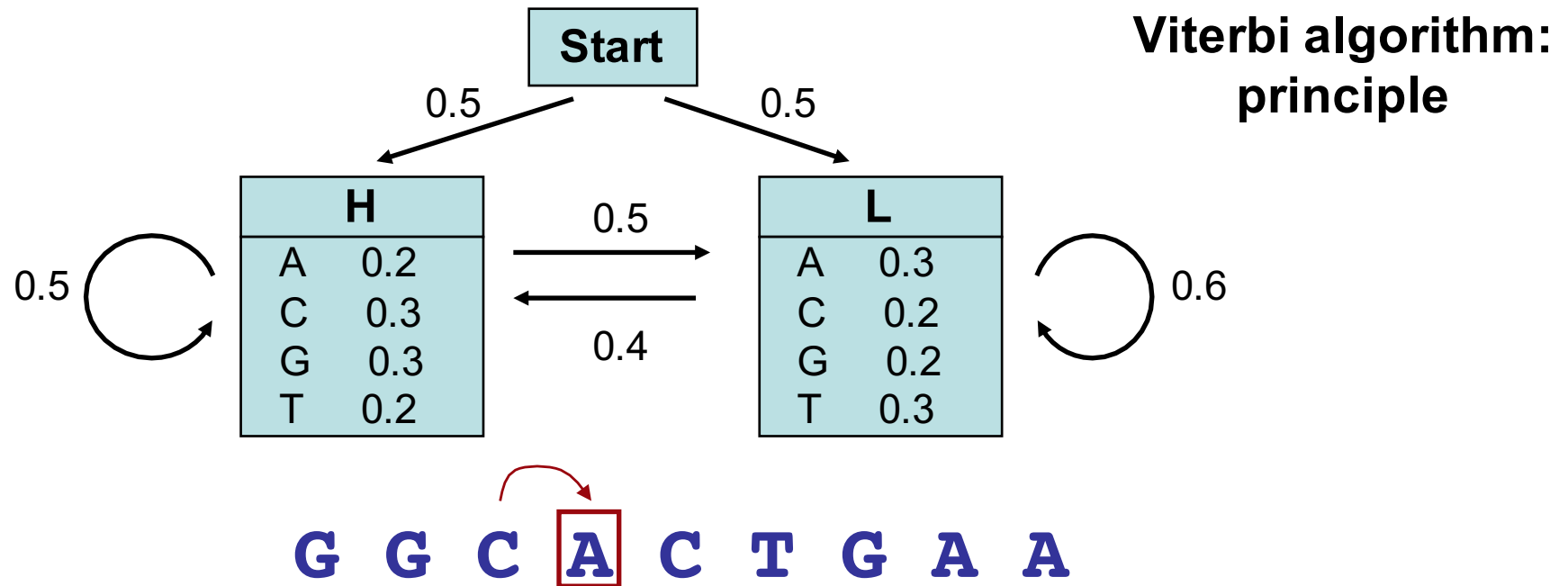


GGCACTGAA

There are several paths through the hidden states (H and L) that lead to the given sequence, but they do not have the same probability.

The **Viterbi algorithm** is a dynamical programming algorithm that allows us to compute the most probable path. Its principle is similar to the DP programs used to align 2 sequences (i.e. Needleman-Wunsch)

HMM : Viterbi algorithm - a toy example



The probability of the most probable path ending in state **k** with observation "**i**" is

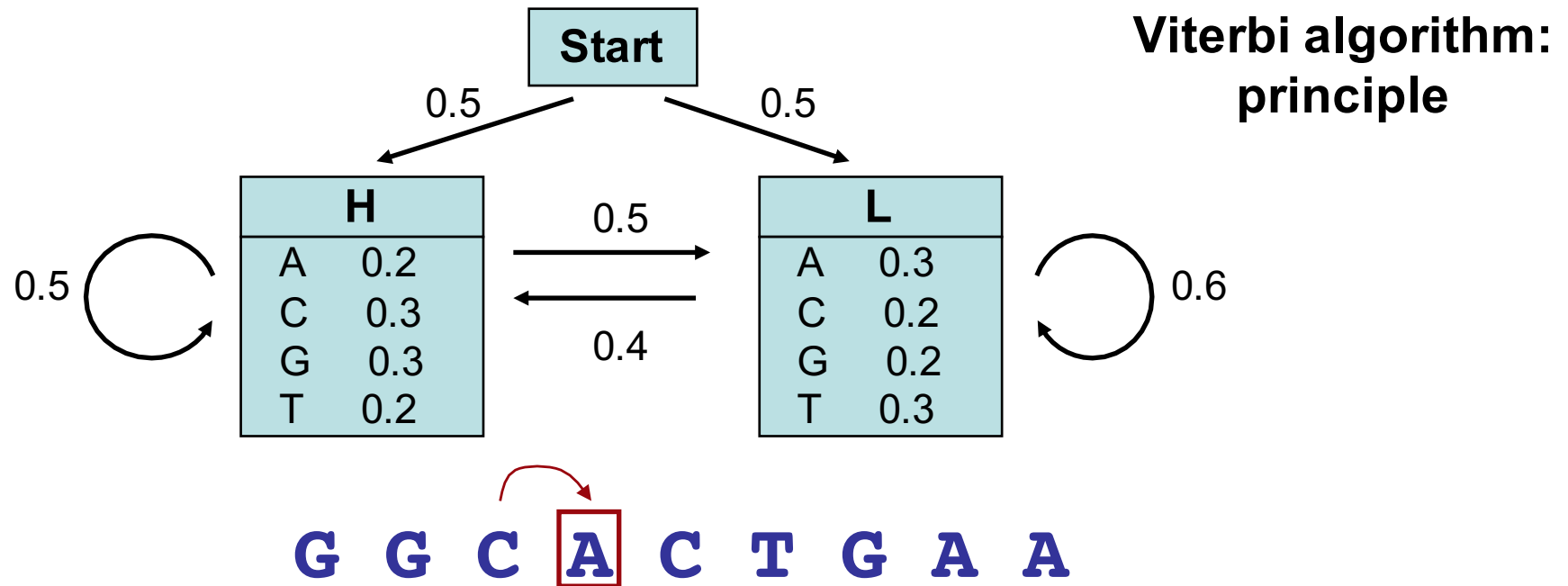
$$p_l(i, x) = e_l(i) \max_k (p_k(j, x-1) \cdot p_{kl})$$

probability to
observe
element *i* in
state *l*

probability of the most
probable path ending at
position *x-1* in state *k*
with element *j*

probability of the
transition from
state *l* to state *k*

HMM : Viterbi algorithm - a toy example



The probability of the most probable path ending in state **k** with observation "**i**" is

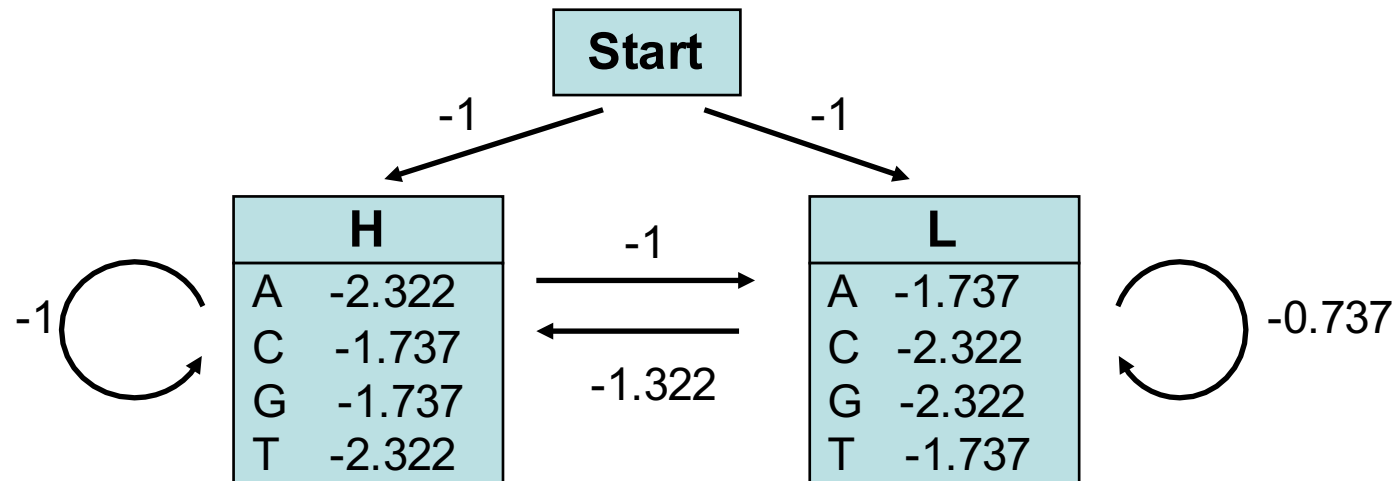
$$p_l(i, x) = e_l(i) \max_k (p_k(j, x-1) \cdot p_{kl})$$

In our example, the probability of the most probable path ending in state **H** with observation "**A**" at the 4th position is:

$$p_H(A, 4) = e_H(A) \max(p_L(C, 3) p_{LH}, p_H(C, 3) p_{HH})$$

We can thus compute recursively (from the first to the last element of our sequence) the probability of the most probable path.

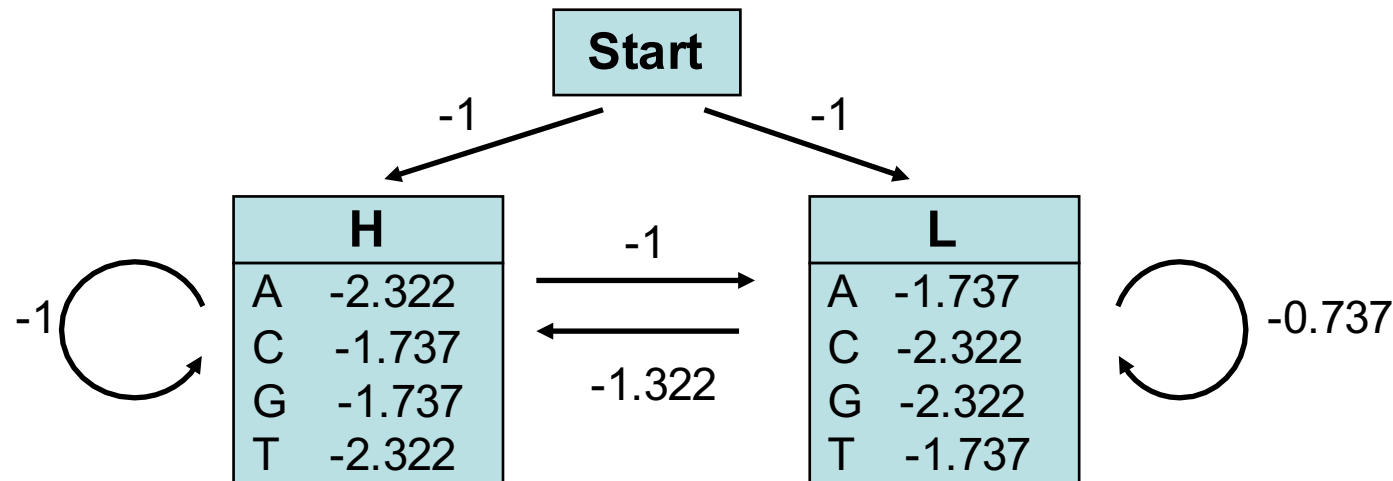
HMM : Viterbi algorithm - a toy example



Remark: for the calculations, it is convenient to use the log of the probabilities (rather than the probabilities themselves). Indeed, this allows us to compute *sums* instead of *products*, which is more efficient and accurate.

We used here $\log_2(p)$.

HMM : Viterbi algorithm - a toy example



GGGCACTGAA

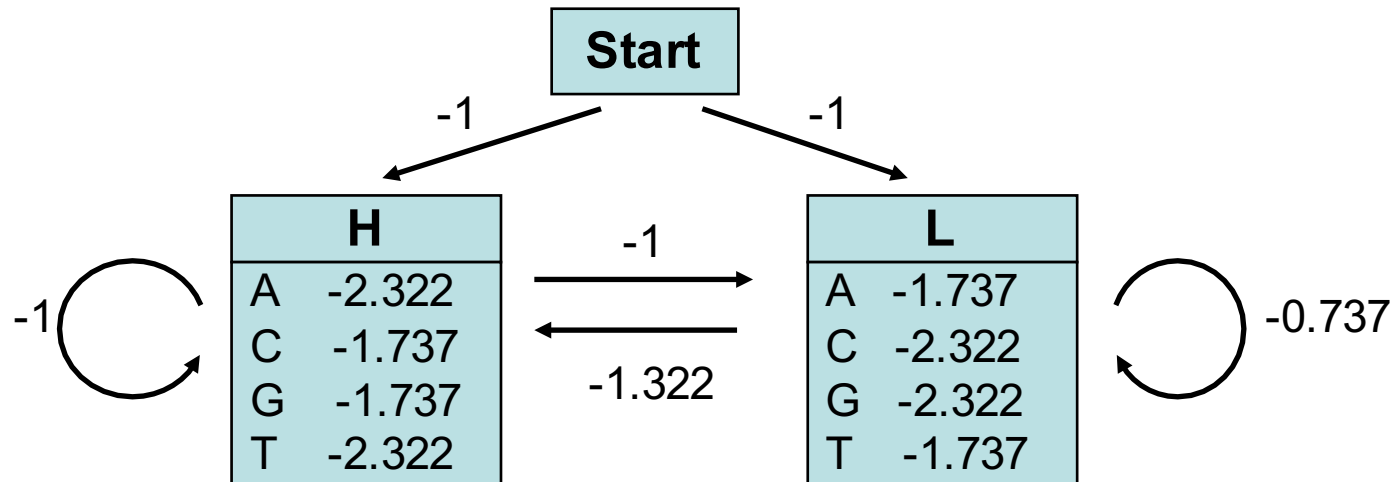
Probability (in $\log_2$) that **G** at the first position was emitted by state **H**

$$p_H(G,1) = -1 - 1.737 = -2.737$$

Probability (in $\log_2$) that **G** at the first position was emitted by state **L**

$$p_L(G,1) = -1 - 2.322 = -3.322$$

HMM : Viterbi algorithm - a toy example



GGCACTGAA

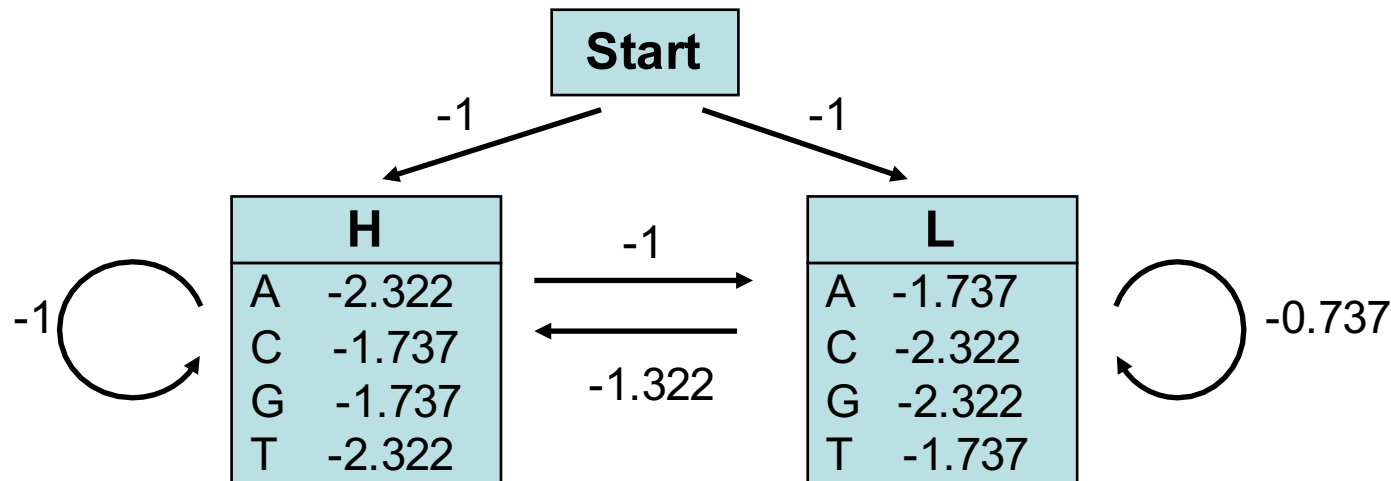
Probability (in $\log_2$) that **G** at the 2nd position was emitted by state **H**

$$\begin{aligned} p_H(G,2) &= -1.737 + \max (p_H(G,1)+p_{HH}, p_L(G,1)+p_{LH}) \\ &= -1.737 + \max (-2.737 -1, -3.322 -1.322) \\ &= -5.474 \text{ (obtained from } p_H(G,1)) \end{aligned}$$

Probability (in $\log_2$) that **G** at the 2nd position was emitted by state **L**

$$\begin{aligned} p_L(G,2) &= -2.322 + \max (p_H(G,1)+p_{HL}, p_L(G,1)+p_{LL}) \\ &= -2.322 + \max (-2.737 -1.322, -3.322 -0.737) \\ &= -6.059 \text{ (obtained from } p_H(G,1)) \end{aligned}$$

HMM : Viterbi algorithm - a toy example

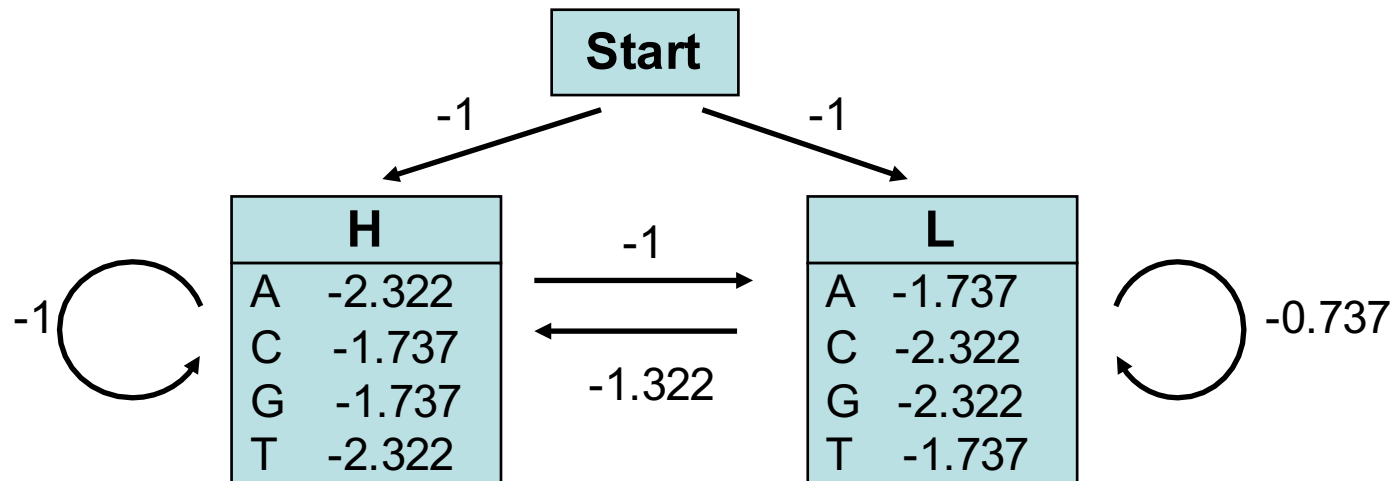


GGCACTGA**A**

	G	G	C	A	C	T	G	A	A
H	-2.73	-5.47	-8.21	-11.53	-14.01	...			-25.65
L	-3.32	-6.06	-8.79	-10.94	-14.01	...			-24.49

We then compute iteratively the probabilities $p_H(i,x)$ and $p_L(i,x)$ that nucleotide i at position x was emitted by state **H** or **L**, respectively. The highest probability obtained for the nucleotide at the last position is the probability of the most probable path. This path can be retrieved by back-tracking.

HMM : Viterbi algorithm - a toy example



GGCACTGAA

back-tracking

(= finding the path which corresponds to the highest probability, -24.49)

	G	G	C	A	C	T	G	A	A
H	-2.73	-5.47	-8.21	-11.53	-14.01	...			-25.65
L	-3.32	-6.06	-8.79	-10.94	-14.01	...	...	...	-24.49

The most probable path is: **HHHLLLLL**

Its probability is $2^{-24.49} = 4.25\text{E-}8$
(remember that we used $\log_2(p)$)